

Indian Institute of Technology Gandhinagar

Financial Modelling and Engineering (ES 418)



Course Project Report

Indian Institute of Technology Gandhinagar

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Algorithmic Trading Strategy Backtesting Platform

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Group 20

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Member Contributions and E-Signatures

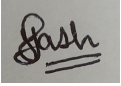
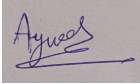
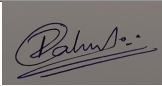
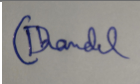
Name	Roll Number	Contribution	E-Signature
Yash Goyal	24110399	Developed the working demo of the project using Streamlit and prepared the complete project report in LaTeX. Also contributed significantly to the project ideation, overall planning, and coding implementation.	
Ayush Tanwar	23110056	Prepared the project presentation and designed the final PPT. Also helped in reviewing the project flow, ideation, and overall concept development.	
Rahul Ramteke	24110288	Contributed to brainstorming and ideation for the project. Helped in collecting information, organizing content, and reviewing different sections of the report.	
Dewansh Singh Chandel	22110072	Assisted in research and gathering references related to the project. Also supported the team in minor documentation work, discussion of ideas, and final editing.	

Table 1: Team member contributions and E-signatures.

1 Introduction

Algorithmic trading refers to the use of computational methods and mathematical models to make trading decisions in financial markets. Instead of relying on manual analysis, trading decisions are generated through predefined rules based on historical data.

In modern financial systems, algorithmic trading has become essential due to its ability to process large datasets efficiently and execute trades with precision. It eliminates emotional bias and enables consistent evaluation of trading strategies.

This project focuses on developing a platform that allows users to design, test, and compare multiple algorithmic trading strategies using historical market data.

2 Motivation and Ideation

Many trading strategies appear effective in theory but may not perform well when applied to real-world data. This creates a need for systematic evaluation methods.

The central objective of this project is:

To evaluate trading strategies in terms of profitability and risk using historical data.

To achieve this, a backtesting platform is developed that simulates trading decisions and provides quantitative performance metrics.

3 Basic Financial Concepts

3.1 Stock Price

The stock price represents the current market value of a company's share. It fluctuates continuously due to demand and supply, company performance, and external economic factors.

3.2 Return

Return measures the percentage change in price over time:

$$\text{Return} = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Returns are used to evaluate profitability and are compounded over time to measure overall performance.

3.3 Risk

Risk refers to uncertainty in returns. Strategies with large fluctuations in returns are considered more risky. Risk is quantified using measures such as volatility and drawdown.

4 Methodology

The system follows a structured workflow:

1. Data is collected using Yahoo Finance.
2. Indicators are computed using historical data.

3. Trading signals are generated based on rules.
4. Signals are applied to simulate trading.
5. Performance metrics are calculated.

All computations are based only on past data, ensuring realistic evaluation.

5 Trading Strategies

In this project, multiple algorithmic trading strategies are implemented and compared. Among them, the Moving Average strategy is considered the **primary strategy**, while the remaining three strategies are used for comparative analysis.

The **primary strategy** refers to the central strategy on which the analysis is focused. It acts as a benchmark for evaluating the effectiveness of other strategies. The additional strategies provide insight into how different trading philosophies perform under identical market conditions.

All strategies are evaluated using historical data of AAPL stock from January 2020 to January 2024.

5.1 Moving Average Strategy (Primary)

The Moving Average (MA) strategy is a trend-following technique widely used in financial markets. It is based on the assumption that prices tend to move in trends, and these trends can be identified using averages of past prices.

A moving average smooths out short-term fluctuations and highlights the underlying direction of the market. In this project, two moving averages are used:

- 50-day moving average (captures short-term trend)
- 100-day moving average (captures long-term trend)

The strategy generates signals based on the crossover of these two averages:

- **Buy Signal:** When the short-term average (MA50) crosses above the long-term average (MA100), indicating the start of an upward trend.
- **Sell Signal:** When MA50 crosses below MA100, indicating a potential downward trend.

This strategy is effective in trending markets, as it captures sustained upward or downward movements. However, it may generate false signals in sideways or highly volatile markets.

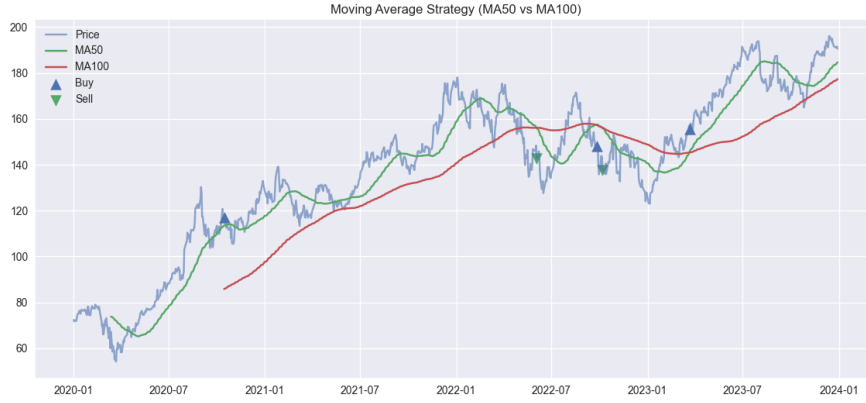


Figure 1: Moving Average crossover strategy applied on AAPL stock (2020–2024).

5.2 Momentum Strategy (RSI)

The Momentum strategy is based on the principle that assets which have performed well in the recent past are likely to continue performing well in the short term.

This strategy uses the Relative Strength Index (RSI), which measures the speed and magnitude of price changes. The RSI ranges between 0 and 100 and is used to identify overbought and oversold conditions.

- $RSI < 30$: Indicates the asset is oversold and may experience a price increase (Buy signal)
- $RSI > 70$: Indicates the asset is overbought and may experience a price decrease (Sell signal)

This strategy attempts to capture short-term reversals and momentum shifts. It is particularly useful in volatile markets but can generate misleading signals during strong trends where prices remain overbought or oversold for extended periods.

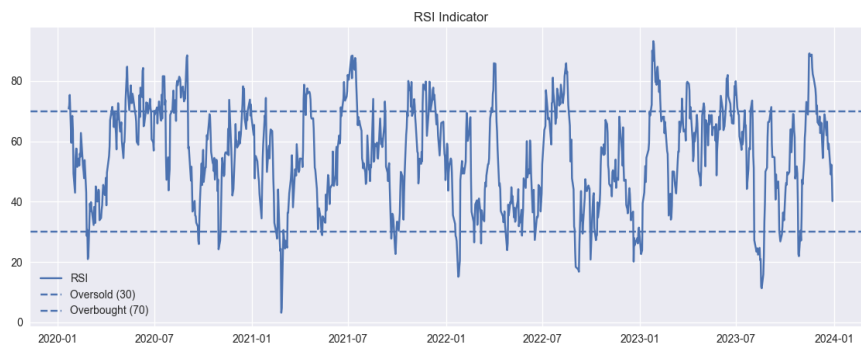


Figure 2: RSI indicator for AAPL stock (2020–2024), highlighting overbought and oversold regions.

5.3 Mean Reversion Strategy

The Mean Reversion strategy is based on the statistical assumption that prices tend to revert to their average value over time. Instead of following trends, this strategy assumes that extreme price movements are temporary and will eventually correct.

A rolling mean is calculated using past price data. The trading logic is as follows:

- If the price is significantly below the mean, it is expected to rise → Buy signal
- If the price is significantly above the mean, it is expected to fall → Sell signal

This strategy performs well in stable or range-bound markets where prices fluctuate around a central value. However, it performs poorly in trending markets, where prices continue to move away from the mean without reverting.

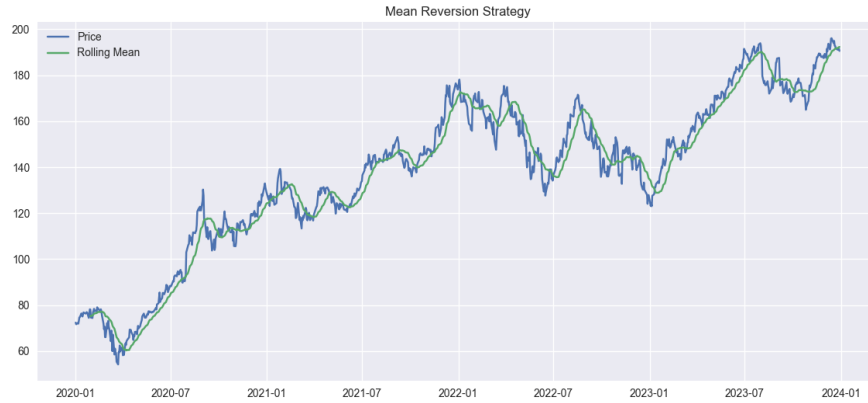


Figure 3: Mean Reversion strategy on AAPL stock (2020–2024), showing deviations from the rolling mean.

5.4 Breakout Strategy

The Breakout strategy is a trend initiation strategy that aims to capture strong price movements when a new trend begins. It is based on the idea that when a price crosses a significant level, it often continues moving in that direction.

In this project, breakout levels are defined using recent highs (resistance) and lows (support):

- **Buy Signal:** When the price breaks above the recent maximum (resistance level)
- **Sell Signal:** When the price breaks below the recent minimum (support level)

This strategy is effective in highly volatile markets where strong trends emerge. However, it is sensitive to false breakouts, where price temporarily crosses a level but does not sustain the movement.

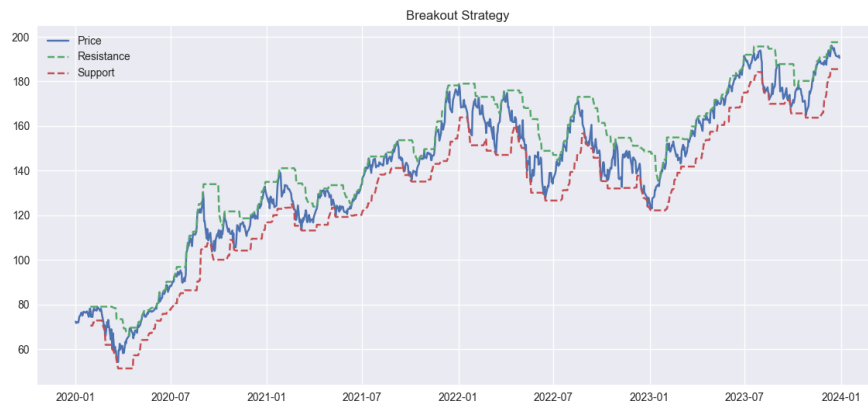


Figure 4: Breakout strategy on AAPL stock (2020–2024), showing resistance and support levels.

6 Backtesting Framework

Backtesting is the process of evaluating a trading strategy using historical data. It allows us to simulate how a strategy would have performed in real market conditions.

At each time step, the system performs the following steps:

- Financial indicators are computed using past data only
- Trading signals (Buy/Sell/Hold) are generated based on the strategy
- These signals are applied to the next time step's return

The strategy return is calculated as:

$$\text{Strategy Return}_t = \text{Market Return}_t \times \text{Signal}_{t-1}$$

This ensures that decisions are made without using future information, making the evaluation realistic and free from look-ahead bias.

Cumulative returns are obtained by compounding daily returns over time, allowing comparison between the strategy and a simple buy-and-hold approach.

Transaction Costs and Realism

In real-world trading, transaction costs such as brokerage fees and slippage affect performance. These costs reduce net returns, especially for strategies that trade frequently.

In this project, transaction costs have not been explicitly incorporated in the implementation. However, they can be modeled by subtracting a fixed percentage (e.g., 0.1% per trade) whenever a buy or sell signal is executed. Incorporating such costs would make the backtesting results more realistic and typically reduce the overall profitability of active trading strategies.

7 Performance Metrics

7.1 Total Return

$$\text{Return} = \frac{\text{Final Value} - \text{Initial Value}}{\text{Initial Value}}$$

Indicates total profitability.

7.2 Sharpe Ratio

$$\text{Sharpe} = \frac{\text{Mean Return}}{\text{Standard Deviation}} \times \sqrt{252}$$

Measures return relative to risk.

Interpretation:

- > 2 : Excellent
- > 1 : Good
- < 1 : Risky

7.3 Maximum Drawdown

$$\text{Drawdown} = \frac{\text{Current Value} - \text{Peak Value}}{\text{Peak Value}}$$

Measures worst loss.

Interpretation:

- < 10%: Low risk
- 10% – 30%: Moderate
- > 30%: High risk

8 Implementation

The proposed system is implemented using Python, leveraging several libraries for data processing, analysis, and visualization. Historical stock price data is obtained using the *yfinance* library, which provides access to financial data from Yahoo Finance.

Data preprocessing and numerical computations are performed using *Pandas* and *NumPy*. These libraries enable efficient handling of time-series data and implementation of rolling window calculations required for indicators such as moving averages, RSI, and statistical measures.

Visualization of results, including price trends and cumulative returns, is carried out using *Matplotlib*. Graphs are generated to illustrate strategy behavior and performance over time. A web-based user interface is developed using *Streamlit*, allowing users to interact with the system. The interface enables users to:

- Select a stock symbol
- Specify a date range for analysis
- Choose a trading strategy
- View performance metrics and graphical results

The integration of backend computation with a user-friendly interface makes the system accessible and easy to use, allowing real-time evaluation of different strategies.

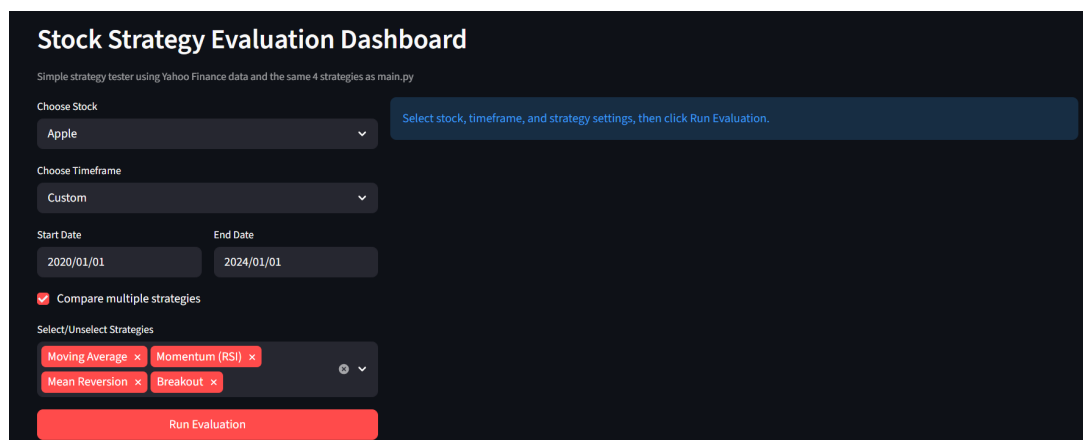


Figure 5: Streamlit-based user interface for selecting stock, date range, and trading strategy, along with visualization of results.

9 Results and Comparison

The performance of all strategies is evaluated using cumulative returns and key financial metrics. Figure 6 shows the comparison of cumulative returns of all strategies against the market benchmark for AAPL stock over the period January 2020 to January 2024. The Market (Buy and Hold) benchmark represents holding AAPL stock continuously over the selected period without active trading.



Figure 6: Comparison of cumulative returns of all strategies and the market benchmark on AAPL stock (2020–2024).

9.1 Performance Metrics Comparison

Table 2 presents a quantitative comparison of all strategies based on total return, Sharpe ratio, and maximum drawdown.

Strategy	Return (%)	Sharpe Ratio	Max Drawdown (%)
Market (Buy and Hold)	150.00	0.85	-31.00
Moving Average	38.53	0.50	-26.58
Momentum (RSI)	1.62	0.11	-27.58
Mean Reversion	-20.31	-0.10	-42.82
Breakout	-30.41	-0.63	-32.21

Table 2: Performance comparison of trading strategies on AAPL stock (2020–2024).

From the results, the Moving Average strategy demonstrates the highest return among all strategies, indicating strong profitability over the selected time period. However, its Sharpe ratio suggests moderate risk-adjusted performance.

The Momentum strategy yields low returns with limited stability, while the Mean Reversion and Breakout strategies result in negative returns, indicating poor performance under the given market conditions.

Overall, the Moving Average strategy proves to be the most effective among the implemented approaches for this dataset.

10 Discussion

The results of the implemented strategies provide important insights into their performance under real market conditions.

A key observation is that the Moving Average Crossover strategy, while being the best among the implemented algorithms with a return of approximately 38.53%, significantly underperformed the Market Benchmark (Buy and Hold), which yielded nearly 150% over the same period.

This difference highlights several important aspects of financial modelling:

- **Trend Lag:** Moving average strategies are inherently reactive rather than predictive. In a strong bull market such as AAPL from 2020 to 2024, the strategy often exits positions during minor corrections and re-enters only after the price has already recovered significantly. This leads to missed gains.
- **Volatility and Whipsaw Effects:** During sideways or uncertain market conditions, frequent crossovers between MA50 and MA100 generate multiple buy and sell signals without a sustained trend. This results in small losses known as whipsaw losses, which accumulate over time.
- **Risk-Adjusted Performance:** The Sharpe Ratio of approximately 0.50 for the Moving Average strategy indicates moderate risk-adjusted returns. This suggests that while the strategy is profitable, the returns are not sufficiently high relative to the volatility experienced.

The Momentum (RSI) strategy showed limited effectiveness, as it struggled to capture sustained trends and was sensitive to short-term fluctuations. The Mean Reversion and Breakout strategies performed poorly in this dataset, primarily due to the strong trending behavior of the market during the selected period.

These observations highlight an important conclusion: no single trading strategy performs optimally under all market conditions. The effectiveness of a strategy depends heavily on the underlying market dynamics.

Furthermore, the comparison with the Market (Buy and Hold) benchmark demonstrates that simple passive investment strategies can outperform active trading approaches, especially in long-term bullish markets.

11 Conclusion

In this project, a complete backtesting platform was developed to evaluate multiple algorithmic trading strategies using historical market data. The system enabled systematic comparison of different strategies in terms of profitability, risk, and overall performance.

Among the implemented strategies, the Moving Average Crossover approach emerged as the best-performing algorithmic strategy, achieving a return of 38.53% with moderate risk characteristics. In contrast, the Momentum (RSI) strategy produced minimal returns, while the Mean Reversion and Breakout strategies resulted in negative performance under the selected market conditions.

However, a key finding of this study is that all algorithmic strategies underperformed the Market (Buy and Hold) benchmark, which generated approximately 150% return over the same period. This highlights an important insight: in strong bullish markets, simple passive investment strategies can outperform active trading approaches.

The results demonstrate that while algorithmic trading strategies provide structured and rule-based decision-making, their effectiveness is highly dependent on market conditions. Trend-following strategies such as Moving Average perform reasonably well in trending markets, but still face limitations such as delayed entry and exit signals.

Overall, this project emphasizes the importance of evaluating both profitability and risk, as well as comparing strategies against a reliable market benchmark. It also highlights that no single strategy consistently outperforms across all scenarios, reinforcing the need for careful strategy selection and evaluation in financial modelling.

12 References

- Website: <https://algotradingstrategybacktest.streamlit.app/>
- GitHub: https://github.com/YashG1836/Algo_Trading_Strategy_Backtest
- Presentation: https://github.com/YashG1836/Algo_Trading_Strategy_Backtest/blob/main/AlgoTrading_Backtesting_Platform.pptx